

6. According to the instructions below, draw the final contents of the register when the command of memory addresses 200 to 206 is executed. (15%)

Instruction format :

opcode (4 Bits)	memory address (12Bits)
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(A)List of opcodes :

opcode	Description	opcode	Description
0000	Load data from memory into register A	0010	Subtract the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

(B)List of opcodes :

opcode	Description	opcode	Description
0000	Load data from memory into register A	0010	Multiply the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

(C)List of opcodes :

opcode	Description	opcode	Description
0000	Load data from memory into register A	0010	Divide the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

Memory initial content :

register (Hex)				
2 0 0				Program counters
				Instruction register
				Register A
				Register B

memory (Hex)	
200	0 4 0 2
202	1 4 0 0
204	2 0 0 0
206	3 4 0 0
.....	
400	0 0 0 2
402	0 0 1 6